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# Large-scale iontronic skins with neural-hierarchical processing architecture

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The human skin gathers and encodes signals from over 200,000 mechanoreceptors, enabling high-spatiotemporal-resolution tactile perception and instant injury prevention. Existing skin-like sensor arrays often have limited sensor count and density, and they use a serial readout architecture that suffers from readout latency, signal crosstalk and bandwidth consumption as element density increases. Here we report a high-density (1,110 units $cm^{-2}$) and large-scale (>10,000 units) neural-hierarchical iontronic skin that mimics the hierarchical processing architecture of the human sensory system, which transmits interactive information in layers via signal encoding, convergence and cognitive modelling. Frequency-based position encoding of iontronic sensors enables parallel signal transmission with suppressed crosstalk and a rate of 100 frames per second, while backend convergent compression decreases bandwidth consumption by hundreds of times. We demonstrate that humanoid robots equipped with neural-hierarchical iontronic skins can perform challenging tasks, including fine tactile interactions that can resolve surface features of 300 μm, and the perception of nociceptive stimuli for self-protective responses. This technology provides a large-scale tactile system with high spatiotemporal resolution, paving the way for humanoid robots to approach human-level perception and manipulation.

The rapid development of artificial intelligence and sensing technologies has propelled remarkable advances in intelligent humanoid robots[1,2]. One of the ultimate goals of human-centred robotics is to achieve human-level perception and manipulation[3,4]. The human skin, the outer covering of the body with an area of ~1.5 $m^2$, contains more than 200,000 mechanoreceptors[5,6]. The myriad mechanoreceptors in the skin can collect tactile information without interferences or latency to provide timely and high-spatiotemporal-resolution sense of touch[7–9]. Replicating such sensory capabilities of the human skin in artificial skins has attracted considerable attentions[10–13]. Although existing sensors can be made to be more sensitive than individual mechanoreceptors, it is challenging to make thin, high-density and large-area flexible sensor arrays that can be conformably deployed on curved surfaces of robots[14,15]. Furthermore, when integrating thousands or more sensors together, the sensory system faces bottlenecks in readout latency, signal crosstalk and bandwidth consumption[16–20].

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Existing artificial skins or sensor arrays mostly use a stacked trilayer structure, together with an array structure with shared row–column electrodes that can effectively simplify wiring for large-scale sensor integration[19–26]. The trilayer sensing structure often consists of two electrodes sandwiching an active layer, for which precise interlayer alignment is required during its fabrication. When scaled up to high-density arrays with thousands of sensing units, alignment error increases with the decreasing feature size of the electrode and the active layer, the increasing sensor count and the enhanced flexibility of the interlayers. As a result, fabrication of such high-density arrays with a trilayer structure is challenging.

In tactile sensor arrays, multichannel signals are commonly read through a serial architecture of time-divisional multiple access[19,21,22], in which individual elements are addressed sequentially and periodically. Such serial readout, however, introduces substantial latency and switching noises[27,28]. The human tactile sensory system encodes information into temporal spikes via primary sensory neurons, which then transmit through parallel neural pathways[6,29]. The posterior cuneate nucleus and spinal cord converge and relay this information to the somatosensory cortex[8]. Pioneering studies have mimicked this biological strategy by converting analogue sensor responses into spike trains[9,27,30–33], using spike frequency or shape to represent tactile features. However, in such systems, each sensing element requires a dedicated circuit to translate signals into spikes, and decoupling the spikes from multiple sensors is challenging. Other studies have proposed frequency-modulated encoding, in which carrier frequency encodes spatial position and amplitude encodes sensing intensity[12,28,34]. Nevertheless, it remains a major challenge to accommodate massive sensors within a limited frequency bandwidth, and to enhance the data transmission efficiency for massive sensors working at high frame rates.

Here, we report a large-scale neural-hierarchical iontronic skin (NI-skin) system that imitates the hierarchical processing architecture of the human tactile nerve bundles. Our system consists of a receptor layer, a spinal layer and a cognitive layer, mimicking the functions of mechanoreceptors, the spinal cord and the human brain, respectively (Fig. 1a). In the receptor layer, iontronic sensing enables high sensing properties and a simplified bilayer sensing structure, supporting high-throughput fabrication of NI-skins with more than 10,000 sensors and a minimum intersensor pitch down to 300 μm (1,110 units $cm^{-2}$). In the intermediate spinal layer, sensing signals are decoded, converged and compressed by hundreds of times to minimize the bandwidth consumption for data transmission. In the cognitive layer, compressed data are received and reconstructed to generate tactile proprioception with high spatiotemporal resolution for robots. This hierarchical hardware–algorithm integration bridges large-scale tactile sensing with robotic cognition, enabling efficient processing of tactile information from more than 10,000 sensing units. We demonstrate multiposition integration of the NI-skins on a robotic platform (Fig. 1b) for applications of fine-texture perception, dexterous manipulation and nociceptive response for self-protection. This work is expected to advance sensing technologies aiming to have high spatiotemporal resolution and large coverage with massive sensors towards human-level robotic perception and manipulation.

## Results

### Design, fabrication and coding principle of the NI-skins

Our NI-skin (the receptor layer) consists of two layers: a coplanar cross-matrix electrode layer with thousands of electrodes (Fig. 2a) and a soft ionogel layer shared by all electrodes. The fabrication of the iontronic sensor array includes three steps (Supplementary Figs. 1–3): the manufacture of the flexible electrode array, the preparation of the microstructured ionogel layer and the assembly of the device. The coplanar cross-matrix electrode was made using a standard flexible printed circuit (FPC) process. The ionogel containing free ions and surface microstructures was prepared via photopolymerization and templating, and further stacked and encapsulated on the coplanar cross-matrix electrode for 'all-in-one' array assembly (Fig. 2b). The microstructured ionogel is shown in the scanning electron microscopy image in Supplementary Fig. 4, and a representative high-density electrode array is shown in Fig. 2c.

In an iontronic sensing element (called 'iontronic receptor'), electronic charges in the electrodes and ionic charges in the ionogel accumulate at the interface to form an electric double layer (EDL) when the ionogel contacts the electrodes[35]. The EDL has a nanoscale charge separation that is orders of magnitude smaller than that in conventional capacitive sensors[36,37], thereby generating strong capacitive signals (Supplementary Fig. 5). The iontronic sensor exhibits a low limit of detection (<0.08 g), minimal drift (9.7% under a 100-kPa load) and stable operation over 100,000 loading–unloading cycles (Supplementary Fig. 6). Furthermore, the iontronic sensors remain responsive over 550 kPa and exhibit a linear response within an operating range of 0–100 kPa (Supplementary Fig. 7). The sensing units can be individually calibrated using unit-specific calibration coefficients, resulting in highly consistent response across all sensing units (Supplementary Fig. 8).

Our device configuration can eliminate the precise interlayer alignment while minimizing wiring complexity in sensor arrays (Supplementary Fig. 9), making it suitable for the fabrication of high-density and large-area sensor arrays. The coplanar cross-matrix electrodes are compatible with commercial FPC processes, supporting scalable manufacturing of sensor arrays (Supplementary Fig. 10). The intersensor pitch of the NI-skin can be made down to 300 μm, corresponding to a sensor density of 1,110 units $cm^{-2}$, exceeding the density of mechanoreceptors in the human fingertip (~240 units $cm^{-2}$)[8,9,32]. Whereas many efforts have been made to enhance the number and density of sensor arrays[5–9,16,21,22,27,28,34], our NI-skins combine a large number of sensors (>10,000 sensing elements), high sensor density (up to 1,110 units $cm^{-2}$) and rapid readout (100 Hz) (Extended Data Table 1). Furthermore, the sensor array has a thickness below 600 μm, which enables a bending radius of 1.2 mm and conformal integration on curved robot surfaces (Supplementary Fig. 11).

In our system, each 'neural bundle' integrates a large number of iontronic receptors (Fig. 2a). To encode spatial positions, each sensing element is excited with a unique frequency, where the frequency domain encodes the location of the iontronic receptor, and the amplitude reflects the intensity of the signals. The method is effective because the EDL capacitance of iontronic sensors is affected by ionic kinetics, with the signal magnitude being frequency-dependent[28]. Figure 2d shows the capacitance–frequency response of our iontronic receptors as a function of applied pressure. In the initial state (before loading), the electrode–ionogel interface lacks intimate contact, and the device behaves as a dielectric capacitor. As pressure increases, the EDL capacitance ($C$) is markedly enhanced, while the signal magnitude decreases as the excitation frequency rises.

To eliminate the impact of frequency, the sensitivity ($S$) of an iontronic receptor is defined as the frequency-normalized change in capacitance with respect to pressure: $S = \delta[(C - C_0)/C_1]/\delta P$, where $C_0$ is the initial capacitance before loading, $C_1$ is the EDL capacitance under saturated pressure, 100 kPa in this study, and $P$ is the instantaneous pressure. The corrected response curves of the iontronic receptors are highly consistent under different excitation frequencies (Fig. 2e). Because of the high magnitude of iontronic capacitance, the sensing performance of iontronic receptors is not affected by the electrical conductance of the substrate (Supplementary Figs. 12 and 13).

In the encoding system, orthogonal frequencies enable parallel transmission of multiple signals within the shared frequency band[38], allowing interference-free separation through fast Fourier transform (FFT). By discretizing the Fourier transform[39], the frequency-bin resolution is expressed as

$$\Delta f_{\mathrm{bin}} = \frac{F_{\mathrm{s}}}{N} = \frac{1}{T_{\mathrm{s}}}, \tag{1}$$

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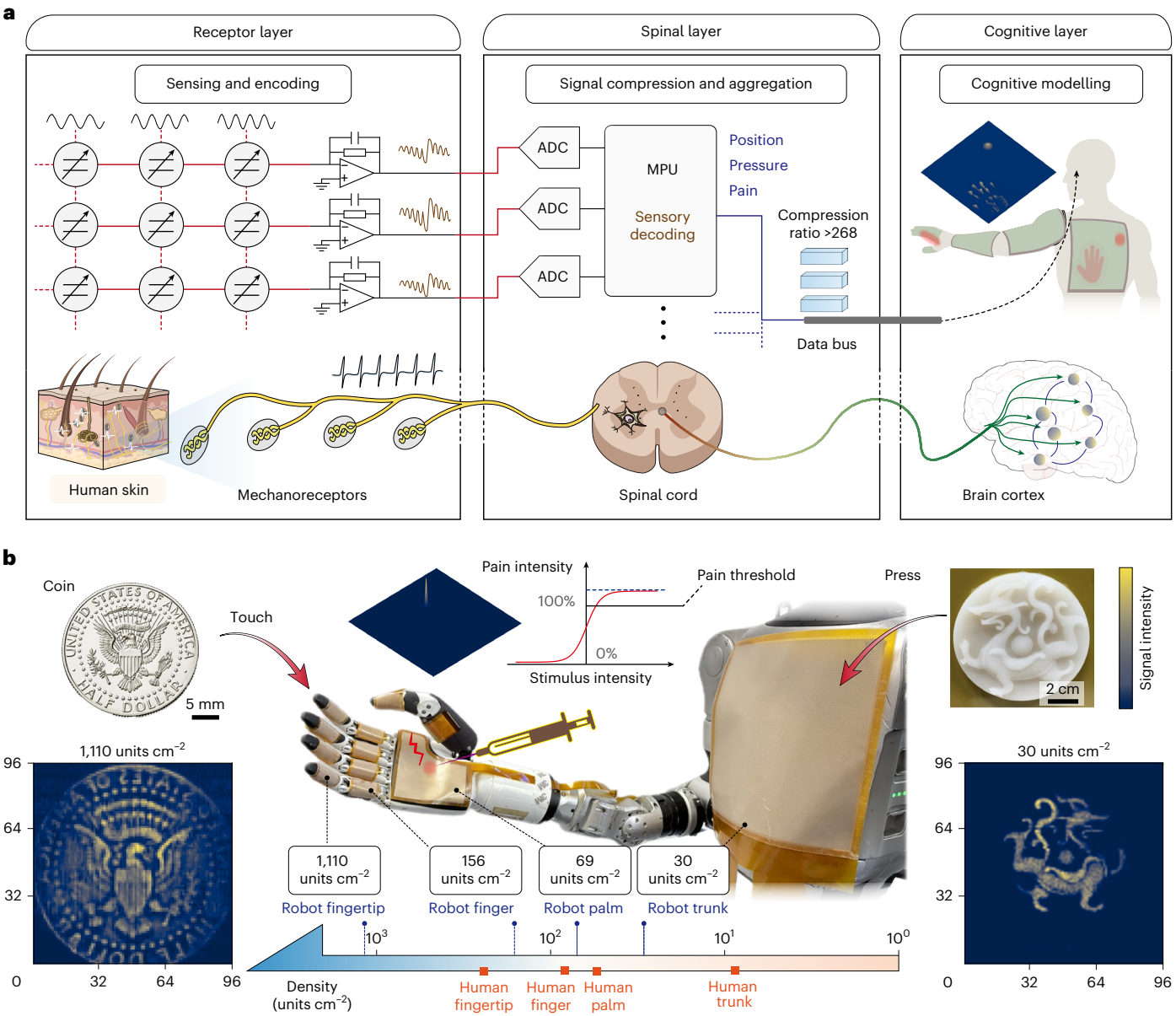



**Fig. 1 | NI-skin system mimicking the hierarchical architecture of the human tactile sensory system. a**, A neural-hierarchical processing architecture comprising three functional layers: a receptor layer, a spinal layer and a cognitive layer. MPU, microprocessor unit. **b**, Applications of the NI-skins with different sizes and different element densities for the detection of fine textures, together with nociceptive sensing for self-protection.

where $F_s$ is the sampling rate of the analogue-to-digital converter (ADC), $N$ is the number of sampling points and $T_s$ is the total sampling period. As $F_s$ decreases (corresponding to an increase in $T_s$), the main lobe of the spectrum becomes narrower, leading to a higher frequency resolution and reduced bandwidth occupation, at a cost of longer sampling and increased processing latency. To ensure the accurate separation of orthogonally encoded frequencies, the encoding frequency interval $f_{inter}$ should satisfy

$$f_{inter} = G\Delta f_{bin}, G \in \mathbb{Z}_{\geq 1}, \quad (2)$$

where $G$ denotes a positive integer ($G \geq 1$) representing the integer multiple of the frequency-bin resolution. Therefore, the frequency selection strategy should balance spectral efficiency against latency and hardware complexity. Here, the sampling period is set to 10 ms, corresponding to a frequency-bin resolution of 100 Hz. To minimize spectral leakage and mutual interference, the encoding frequency interval is set to 200 Hz. Within each neural bundle, 96 neurons are assigned to an encoding frequency band ranging from 1 to 20 kHz (encoding circuit is shown in Supplementary Fig. 14). Five independently fabricated iontronic sensing elements were characterized at 96 encoded excitation frequencies ranging from 1 to 20 kHz (Fig. 2f). The normalized sensitivity is defined as $S/S_{mean}$, where $S_{mean}$ is the mean sensitivity across all excitation frequencies. Across all tested sensing elements, the normalized sensitivity was 1.000 ± 0.006 (mean ± s.d.). This result ensures consistent responses of all iontronic sensors across different encoding frequencies.

In a neural bundle, the frequency encodes the spatial location of the receptors, while the amplitude reflects the magnitude of the stimulus. Mechanical stimulation on different sensing elements of the NI-skin produced specific encoding frequency responses, with the output current amplitude varying with pressure (Fig. 2g,h). As shown in Fig. 2h, when pressure is applied to the iontronic receptors encoded at 4.0, 4.2 and 4.4 kHz, the neural bundle detects the corresponding

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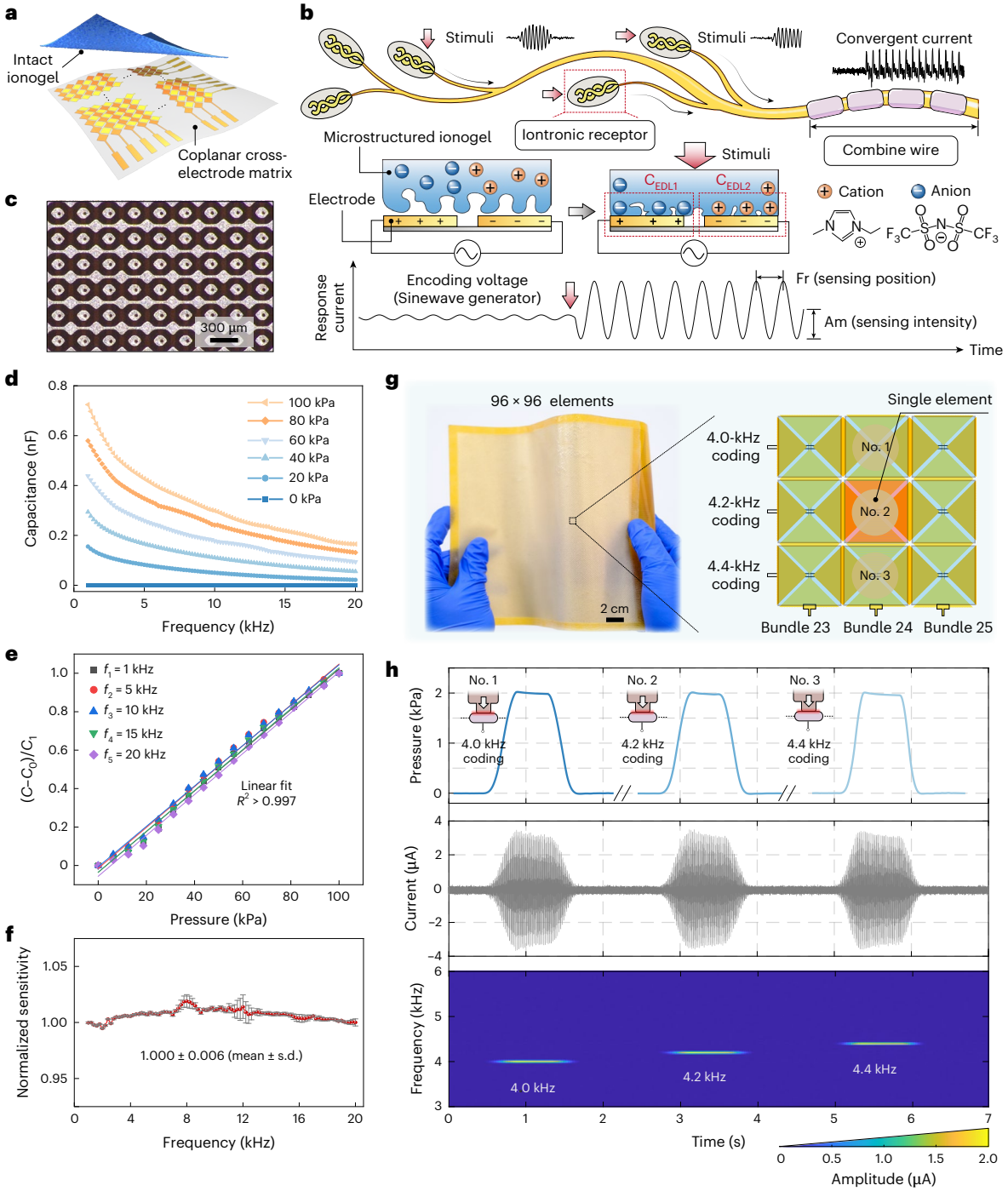



**Fig. 2 | Design and coding principle of the NI-skin system. a**, A schematic illustration of the NI-skin with a bilayer sensor architecture. **b**, A schematic illustration of the structure and sensing principle of iontronic receptors. **c**, An optical image of high-density electrode array. **d**, Capacitance response versus excitation frequency under varying applied pressures. **e**, Frequency-normalized response curves of the sensing element at different excitation frequencies (*f*). **f**, Consistency of normalized sensitivity across 96 encoded excitation frequencies from 1 to 20 kHz. Data are mean ± s.d. from five independently fabricated sensing elements ($n = 5$). **g**, A photograph of the large-scale 96 × 96 NI-skin and schematic illustration of the frequency-encoded signals used for spatial localization at the receiver. **h**, Time-frequency analysis of encoded signal response under a constant load at different positions. Fr, frequency; Am, amplitude.

time-frequency responses and output current pulses with clear feature frequencies. This capability enables parallel signal perception and transmission from a large number of receptors.

## Decoding and convergence of large-scale signals

In the human sensory system, the mechanoreceptors in the lower receptor layer can respond to touch and transmit signals to the backend through nerve bundles[32,40]. The posterior spinal cord is responsible for signal relay and preliminary processing[41], ensuring that aggregated information is efficiently transmitted to the brain while simultaneously enabling rapid initiation of protective reflexes. This neural-hierarchical processing architecture enables large-scale tactile information from the numerous mechanoreceptors to be perceived and transmitted to the neural system. Inspired by this, we designed a 'spinal layer' to decode

sensing signals and perform compression-based preprocessing (Fig. 3a) that generates aggregated signals to the 'brain' (central processor).

In this signal transmission framework, each neural bundle can be equipped with up to 96 iontronic receptors, encoded by frequency signals ranging from 1 to 20 kHz with a 200-Hz interval. The current signals of the receptors are superimposed and converted through a current-to-voltage converter (C/V converter), followed by acquisition via an ADC (Supplementary Figs. 15a,b and 16). According to the Nyquist sampling criterion, the ADC sampling rate is set to 51.2 kHz, which is at least twice the bandwidth of the encoded signal, ensuring alias-free observation for FFT-based decoding. The FFT algorithm is executed on a digital signal processor block to reconstruct the location and intensity information from the frequency-encoded signals (Supplementary Fig. 15c). As shown in Fig. 3b–d, when a pressure of 10 kPa is applied to receptors at different locations, the signals output by the C/V converter can be decoded without frequency aliasing. As a result, the adjacent leakage ratio (ALR) between encoded frequencies, defined as the ratio of the signal response leaked into adjacent frequency channels to that of the main spectral component, remains below 1%.

In the excitation–detection scheme, frequency-encoded voltages are applied to the excitation electrodes, while the sensing electrode converges the corresponding currents, enabling capacitance readout through frequency-domain decoding[28] (Supplementary Note 1). A 512-point sampling scheme ($\Delta f_{bin}$ = 100 Hz) was selected as the optimal operating point to balance frequency discrimination, decoding latency and power consumption, yielding a per-frame readout time of 10 ms (Supplementary Figs. 17 and 18). The system can be driven by a portable battery lasting for at least ~12 h (Supplementary Fig. 19), supporting sustained sensing in a robotic system. The potential increase in energy consumption with the increasing number of sensors and readout frame rate can be addressed by adaptive frame-rate scheduling, which matches the readout rate to task requirements, together with event-driven readout.

Conventional iontronic sensor arrays using a piece of shared ionic material may experience substantial crosstalk between sensing elements due to ion migration[22]. As shown in Fig. 3e, the leaked electric field induces ion migration, resulting in EDL formation at surrounding sensing units—an undesired crosstalk. To suppress this effect, we designed a parallel zero-potential (PZP) scheme at the decoding end, where all sensing electrodes are maintained at an equivalent zero potential (Fig. 3f). During readout, each sensing unit is addressed at the intersection of the excitation and sensing electrodes. The PZP configuration confines the excitation electric field to the addressed sensing node, thereby preventing ion migration beyond the electrode intersection. As a result, even with a layer of shared ionogel, EDL capacitance is formed exclusively at the addressed sensing node, effectively eliminating ion-migration-induced crosstalk. In addition, the PZP mechanism can interrupt the loop current between the parallel components (Fig. 3g), effectively suppressing electrical crosstalk within the parallel circuits. This dual mechanism simultaneously mitigates ionic and electrical crosstalk while enabling parallel current summation from multiple sensing elements.

Our experimental data show that this method reduces the crosstalk of the NI-skin from ~39.4% to ~1.3%, which is even smaller than that of the scheme with isolated ionogel for each sensing element (~1.7%) (Fig. 3h). Moreover, the proposed architecture suppresses crosstalk to below 1.38% across NI-skins with different array sizes, and crosstalk suppression remains effective on both conductive and insulating substrates (Supplementary Figs. 20 and 21). Our approach enables crosstalk-free and high spatial-resolution sensing, enabling the array to resolve surface features as small as 300 μm (Supplementary Figs. 22 and 23 and Supplementary Videos 1–4).

A compression block based on three-dimensional discrete wavelet transform (3D-DWT) is implemented in the spinal layer to simplify data transmission (Supplementary Fig. 24). The wavelet transform enables highly sparse and compact representations and can be efficiently realized in an edge-computing hardware[42]. The 3D-DWT block decomposes the 3D signals into sparse frequency domain coefficients, which are then quantized to reduce the data size. Finally, entropy encoding exploits statistical redundancy to further compress the quantized coefficients. The 3D-DWT-based compression aims to effectively reduce the data transmission size while preserving tactile features with minimal acceptable accuracy loss.

## Data compression-enabled lightweight tactile cognition modelling

High spatiotemporal-resolution signals from numerous mechanoreceptors of the human skin are transmitted to the cerebral cortex layer via a data bus for cognitive analysis and perceptual construction[43]. We constructed a robotic sensory system based on the NI-skins and a hierarchical processing architecture to mimic this process, as shown in Fig. 4a. The NI-skins provide massive data, and the hierarchical architecture enables rapid processing and signal transmission of the data.

In the hierarchical architecture, the intermediate spinal layer is responsible for aggregating and compressing data, with the degree of compression quantified by the compression ratio (CR), defined as the ratio between the size of raw data to that of compressed data. In the 3D-DWT block (Supplementary Fig. 24), a larger quantization factor $Q$ results in a higher CR at the expense of potential data distortion (Fig. 4b). At $Q = 2,000$, the data are sparsified to achieve a CR value of 268 while preserving the spatial structure (structural similarity index measure (SSIM) > 0.97) and fine temporal features (Supplementary Fig. 25).

A robot is capable of extracting 3D spatiotemporal features from sensing signals to perform the recognition and cognition of objects being touched[44]. Here, we designed a 3D convolutional neural network (3D-CNN; Supplementary Fig. 26) to extract spatiotemporal features from the captured touch signals. This process can reduce the transmission bandwidth consumption from 14.7 megabits per second (Mbps) to 56.7 kilobits per second (kbps) (Fig. 4c). Using the compressed sensing strategy, sparse features enable inference with a lighter network. Compared with the original neural network ($3.95 \times 10^9$ floating-point operations (FLOPs) and 80,000 parameters), the lightweight network requires only $9.38 \times 10^8$ FLOPs (23.7% of the original) and 3,600 parameters (4.5% of the original).

Our strategy maximizes the CR value while retaining essential features. The 3D-CNN extracts spatiotemporal features, with the $t$-distributed stochastic neighbour embedding ($t$-SNE) projection illustrating data clustering. The Davies–Bouldin index (DBI) was used to assess cluster separability, with a lower DBI indicating better intraclass compactness and interclass distinction, validating feature discriminability. We constructed a dataset for grasping ten object categories (Fig. 4d, grasp dataset, and Supplementary Fig. 27a), capturing responses during manipulation. Increasing CR to 268, the clustering quality of 3D-CNN-extracted features slightly decreases (DBI of 1.17), yet recognition accuracy remains above 98.2% (Fig. 4e). A dataset (Fig. 4f, touch dataset, and Supplementary Fig. 27b) involving ten touch gestures was further created to capture dynamic interactional responses. At a CR of 268, 3D-CNN-extracted features remain well-clustered (DBI of 0.89), with a recognition accuracy above 95.4% (Fig. 4g).

Further increase of CR to an aggressive value of ~460 will reduce the recognition accuracy, especially for touch gesture recognition, which is more sensitive to aggressive compression and exhibits substantial interclass confusion (Supplementary Figs. 28 and 29). Note that object recognition tasks are more insensitive to compression. A CR of 268 is an optimal value because it achieves high accuracy for both the recognition of objects and gestures, while effectively reducing the transmission time from 208 ms for the case without compression to <1 ms (Supplementary Fig. 30). In fact, the complete latency breakdown of the full pipeline is longer—the sampling,

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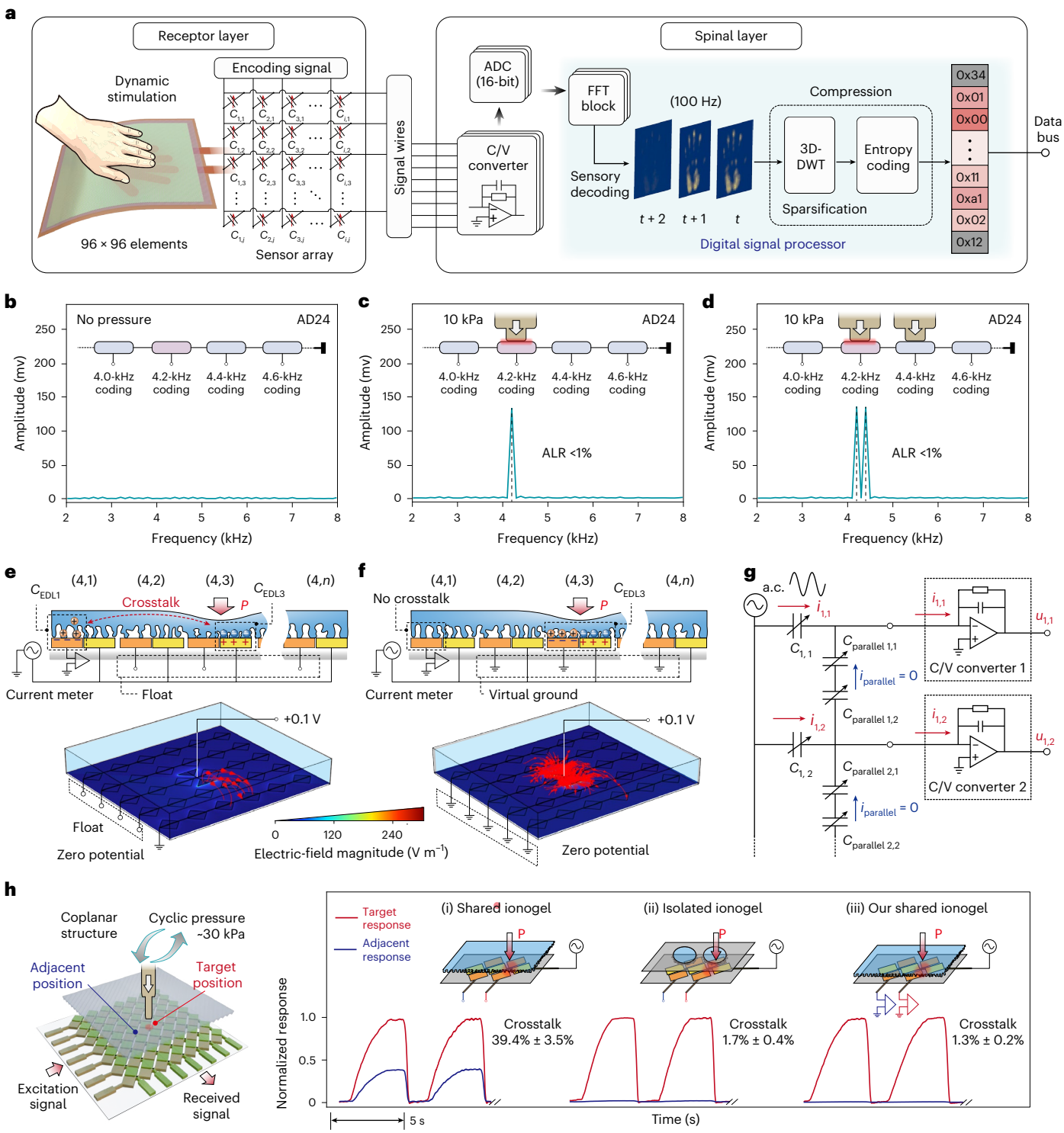



**Fig. 3 | Decoding and convergence of large-scale signals. a**, A flowchart of decoding and compressed convergence for sensory-encoding signals. The spinal layer is responsible for decoding encoded signals from receptors, subsequently compressing and aggregating the signals onto a data bus for the transmission to the central processor. **b**–**d**, Response of decoded signals under pressure stimuli at different locations. The ALR quantifies the extent to which the energy of the main spectral peak leaks into adjacent frequency bins. **e**, A schematic illustration (top) and simulation (bottom) of ion-migration-induced crosstalk under electrode floating connection. **f**, A schematic illustration (top) and simulation (bottom) of ion-migration-induced crosstalk suppression in the NI-skin decoding system equipped with the PZP mechanism. **g**, A schematic illustration of electrical crosstalk suppression between parallel circuit elements in the array under the PZP mechanism. *i*, line current; *u*, node voltage. **h**, Crosstalk suppression analysis between iontronic sensing elements. (i) Without crosstalk suppression, the crosstalk reaches up to ~39.4%. (ii) Under ionogel separation, crosstalk can be reduced to be ~1.7%. (iii) With our PZP design, crosstalk is reduced to ~1.3%.

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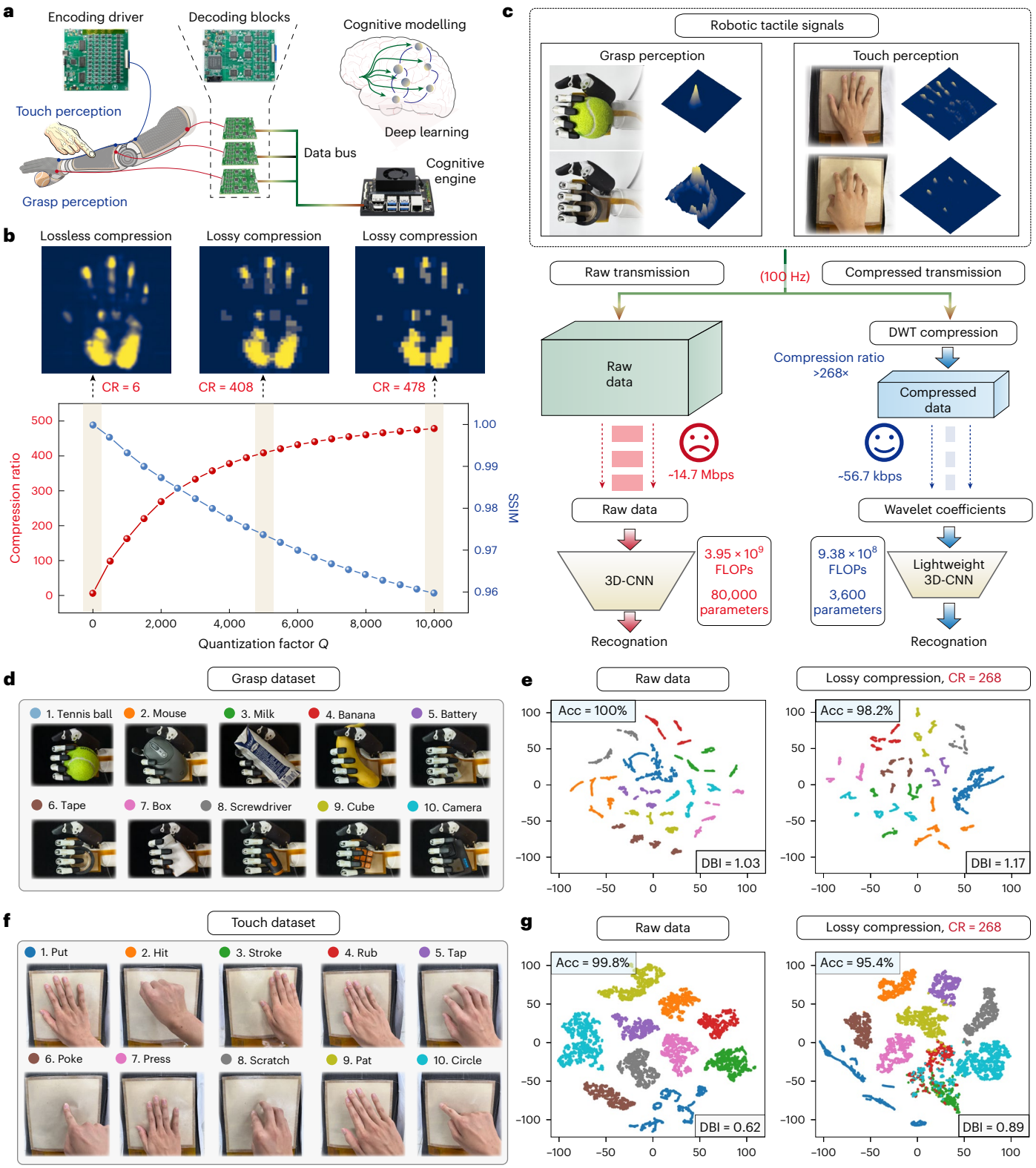



**Fig. 4 | Data compression for lightweight robotic tactile cognition modelling.** **a**, Hardware framework of the hierarchical processing architecture for the aggregation and transmission of large-scale tactile information. **b**, Trade-off between CR and SSIM for optimizing lossy compression performance. **c**, Demonstration of the advantages of tactile compression in reducing transmission bandwidth and enabling lightweight inference. **d**, Dataset of grasping ten different objects. **e**, *t*-SNE visualization of the grasp dataset shows clustering of raw and compressed tactile events, indicating that key palmar tactile features are retained at high CRs. Acc, accuracy. **f**, Dataset of ten distinct touch actions. **g**, *t*-SNE visualization of the touch dataset reveals clustering of raw and compressed tactile events, demonstrating that key somatosensory features are preserved even at high CRs.

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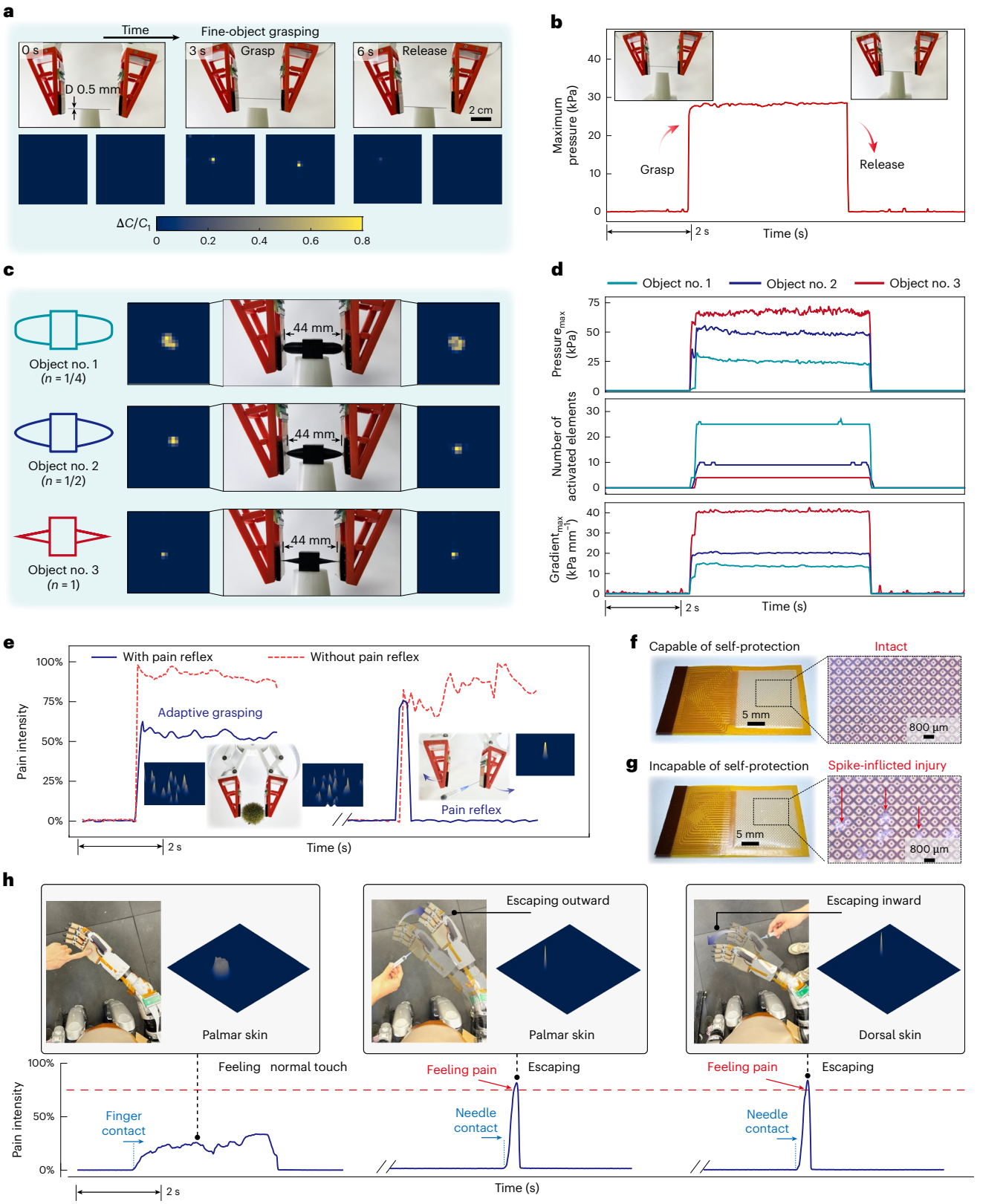



**Fig. 5 | Robotic fine manipulation and self-protection enabled by high-spatiotemporal-resolution tactile perception. a**, A demonstration of a robotic hand equipped with NI-skin grasping a pencil lead with a 0.5-mm diameter. *D*, diameter. **b**, A real-time response curve of the maximum pressure on the NI-skin during a robotic finger grasping the pencil lead. **c**, Tactile response distribution of high-density NI-skin during robotic finger grasping of objects with varying sharpness, modulated by the power-law index *n* (nos. 1–3). **d**, Tactile feature responses of high-density NI-skin during robotic finger grasping of objects with varying sharpness, including maximum pressure, activated element count and maximum pressure gradient. **e**, Demonstration of safe interaction and self-protection with a robotic finger equipped with high-spatiotemporal-resolution tactile perception. **f**, Effectiveness of robotic self-protection with tactile pain perception. **g**, Mechanical injury caused by the absence of pain perception and timely avoidance in robots. **h**, Demonstration of pain-triggered self-protective responses in an embodied robot equipped with NI-skins.

FFT-based demodulation, compression, transmission and inference totals ~23 ms (Supplementary Fig. 31), which is still sufficiently short to support real-time robotic interaction. Moreover, the proposed framework demonstrates good generalization to previously unseen objects and remains applicable to multicontact tactile cognition (Supplementary Figs. 32 and 33).

### Manipulation of fine objects

Human hands can dexterously manipulate small and fragile objects, which requires tactile perception with high spatiotemporal resolution. Here, we developed a soft robotic finger equipped with NI-skins that can detect stimuli from sharp objects with a two-point limen of 0.8 mm (Supplementary Fig. 34), surpassing the smallest two-point limen (down to 2 mm) of a human fingertip[32]. As shown in Fig. 5a and Supplementary Video 5, the NI-skin can reliably record real-time pressure distributions when robotic fingers grasp a 0.5-mm-diameter pencil led, and the corresponding interaction signals are shown in Fig. 5b. Utilizing our tactile system, the fingers can delicately manipulate an even thinner object, a soft solder wire with a diameter of 0.2 mm, without inducing buckling when an appropriate load is applied by compressing the wire at both ends (Supplementary Video 6). When buckling is induced by a human hand, the robotic fingers rapidly adjust their positions so that the wire does not fall off.

### Nociceptive sensing and self-protection based on pain feedback

An important capability of human skin is the sensing of harmful pain to trigger rapid self-protection against puncture[45–48]. Our tactile system aims to replicate this capability by quickly detecting nociceptive stimuli at the spinal layer, triggering a protective response. Nociceptive stimuli often arise from interactions with sharp and stiff objects (for example, needles) that produce a high pressure gradient, $\Delta P/\Delta x$, which represents the change in pressure ($\Delta P$) over a distance ($\Delta x$). The minimum value of $\Delta x$ is often determined by the two-point limen. Here, the NI-skin has a small two-point limen down to 300 µm, enabling detection of sufficiently high $\Delta P/\Delta x$ values. Stimuli with varying sharpness levels induce different sensations of pricking on the skin. We thus selected target objects with different sharpness to stimulate the skin of the finger. The object curvature follows a power law, where the radius of curvature at the edge of the protruding feature can be modified by the power law exponent $n$ (refs. 45,49) ($0 \le n \le 1$; model formula in Supplementary Note 2 and Supplementary Fig. 35a). As shown in Fig. 5c, increasing $n$ corresponds to greater edge curvature and a sharper object, leading to a smaller response area of the NI-skin during finger grasping. During grasping and releasing cycles, the NI-skin monitored key features of maximum pressure, activated element count and maximum pressure gradient (Fig. 5d). As $n$ increases, maximum pressure and pressure gradients increase, while the number of activated elements decreases. These features were used for nociceptive sensing.

We stimulated the NI-skin on the robotic finger with spikes of varying sharpness to quantify the intensity of 'pain'. We applied stimuli with objects at different $n$ values, recording NI-skin responses to create a pain-perception dataset (Supplementary Fig. 35b). A pain prediction model based on logit regression was developed using three features: maximum pressure, the number of activated sensing elements and the maximum pressure gradient, to assess pain intensity (Supplementary Note 3 and Supplementary Fig. 36). On the pain-perception dataset, the model achieved a coefficient of determination ($R^2$) up to 91.1%.

The pain threshold ($P_{th}$), defined as the critical value output in the pain model that triggers self-protection, is used to prevent further damage to the robot. We show that the robotic hand with its two fingers equipped with the NI-skins performs protective reaction as it manipulates a spiny chestnut (Fig. 5e and Supplementary Video 7). In this case, a large number of sensing elements are activated. While at least one sensing element of the NI-skin is subjected to $P_{th}$, the fingers are triggered to decrease the grasping pressure for the protection of the skin. In another scenario, when the fingers of the robotic hand touch a needle, only one sensing element on the skin is activated with a high pressure gradient exceeding its $P_{th}$, triggering a protective response (Fig. 5e and Supplementary Video 7). This case is judged as the needle puncture, and the fingers are thus controlled to withdraw rapidly to prevent further injury of the NI-skin (Fig. 5f). Without pain perception, however, the fingers continue moving towards the needle, causing damage to a few sensing areas of the NI-skin (Fig. 5g). We demonstrated an embodied robot equipped with the large-area NI-skins that detect nociceptive stimuli to trigger self-protective responses (Fig. 5h and Supplementary Video 8), highlighting the potential of the NI-skins to enhance both dexterous manipulation and self-protection in embodied robotic systems.

## Discussion

A key goal in human-centred robotics is to achieve human-level perception and manipulation with fine interaction and self-protection. By integrating frequency-encoded receptors, crosstalk suppression and compressed sensing, the NI-skin mimics the hierarchical processing of the human tactile sensory system, which comprises receptor, spinal and cognitive layers. This architecture achieves a minimum intersensor pitch of 300 µm and crosstalk-free readout of over 10,000 sensing elements at 100 frames per second. Its ability to manipulate fragile objects and provide real-time nociception demonstrates its potential to enhance robotic dexterity and safety.

This work may provide guidelines for the design of next-generation tactile perception architecture suitable for large-scale robotic skins. The current implementation of the encoding–decoding circuitry is relatively bulky, limiting miniaturization and full integration of the readout interface into artificial skin. Future work will explore integrating the interface circuitry into application-specific integrated circuits, with our design serving as an effective architecture for such application-specific integrated circuit chips.

## Methods

### Materials

The following chemicals were purchased from Aladdin: 3,6-dioxa-1,8-octanedithiol (D154345), 1,4-butanediol divinyl ether (B472435), 1,1,1-tris(3-mercaptopropionyloxymethyl)propane (T464015), 2,2-dimethoxy-2-phenylacetophenone (B105178), 1-ethyl-3-methylimidazolium bis(trifluoromethylsulfonyl)imide ([EMIM][TFSI]; E101506). The thermoplastic polyurethane (TPU) films were purchased from Dongguan Lijin Polyurethane. The microstructured frosted glass was purchased from Anhui Hengboli Instrument Technology.

### Fabrication of the NI-skin

The NI-skin was fabricated in three steps (Supplementary Figs. 1–3): fabrication of the flexible high-density electrode array, preparation of the microstructured ionogel sensing layer and vacuum thermo-compression-assisted assembly of the sensor.

First, the flexible high-density electrode arrays were fabricated using a standard commercial FPC process. The process included drilling, plated through-hole formation, photolithographic patterning, chemical etching, polyimide coverlay lamination and surface finishing. The customized electrode arrays were manufactured by Shenzhen JLC Technology Group and Shenzhen Lianzhongbang Electronics, China.

Second, the microstructured ionogel sensing layer was prepared via photopolymerization. The ionogel precursor was composed of 2.7345 g (15 mmol) of 3,6-dioxa-1,8-octanedithiol (dithiol monomer), 2.9862 g (21 mmol) of 1,4-butanediol divinyl ether (divinyl monomer), 1.5942 g (4 mmol) of 1,1,1-tris(3-mercaptopropionyloxymethyl)propane (trithiol crosslinker), 0.036 g (0.5 wt% relative to total monomer) of 2,2-dimethoxy-2-phenylacetophenone (photoinitiator) and 0.5852 g

(8 wt% relative to total monomer) of 1-ethyl-3-methylimidazolium bis(trifluoromethylsulfonyl)imide (ionic liquid). The precursor solution was thoroughly mixed, casted onto a microstructured mould, laminated with a TPU film, and cured under UV irradiation. The resulting microstructured ionogel/TPU layer was then peeled off from the mould and cut to desired dimensions.

Third, the microstructured ionogel layer was bonded to the flexible electrode array using commercially available double-sided adhesive tape, which served as both an adhesive layer and a spacer. The assembled device was subsequently encapsulated by vacuum thermo-compression, integrating the ionogel film with the high-density electrode array to form the NI-skin.

### Characterization of sensing properties of sensing elements

An inductance–capacitance–resistance meter (E4980AL, KEYSIGHT) was initially used to measure the capacitance of the sensors. Capacitance measurements were all performed at 100 mV and in the frequency range from 100 Hz to 100 kHz. The 3D-printed indenters were used to apply the displacement loads. The pressure applied to the sensors was controlled and recorded by a tensiometer machine (LD23.501, LSD). A lock-in amplifier (MFLI, Zurich Instruments) was used for aliasing signal acquisition and spectrum analysis. All data plots presented in the Article and the Supplementary Information were processed with Origin 2018 (OriginLab Corporation).

### Development of interface circuitry for NI-skin

The design of the encoding driver is shown in Supplementary Fig. 14. Direct digital synthesizers (AD9833, Analog Devices) were used to generate frequency-tunable a.c. excitation signals. An STM32F103V8T6 microcontroller (STMicroelectronics) was used to control the encoding driver and configure the direct digital synthesizer devices. The backend circuit consisted of a high-pass filter with a cutoff frequency of 200 Hz and a voltage follower implemented using an operational amplifier (OPA4388, Texas Instruments). The microcontroller firmware was developed using STM32CubeMX version 6.14.0, STM32CubeIDE version 1.18.0 (STMicroelectronics) and Keil MDK-ARM version 5.39.

The design of the decoding block is shown in Supplementary Fig. 16. The C/V converters were implemented with a precision operational amplifier (OPA4388, Texas Instruments), in which the reference resistor was realized with a programmable digital potentiometer (TPL0501-100) to enable adaptive adjustment of the output conversion voltage. In addition, 16-bit ADCs (AD7606, Analog Devices) were used to acquire voltage signals from the analogue frontend, with each ADC chip providing eight synchronous voltage-sampling channels. The sampling rate was set at 51.2 kSPS, and the data were read out over a parallel digital interface. A field-programmable gate array (FPGA; Altera EP4CE10E22C8N, Cyclone IV E) was deployed for parallel acquisition from multiple ADCs and subsequent parallel FFT demodulation. The FPGA performed multichannel parallel ADC sampling and parallel FFT demodulation (512-point) using digital signal processor blocks to extract the frequency-encoded sensory response. The demodulated signals were then aggregated and compressed, and the compressed data were transmitted to the central processor over USB or Ethernet, which served as the main data bus. The FPGA logic was designed and compiled using Quartus II version 13.0.

### Demonstration setup

NI-skins with different sensor counts and sensor densities were developed for tactile perception and self-protection in embodied robotic platforms. A high-density fingertip NI-skin (96 × 96 elements; 1,110 units $cm^{-2}$) was used to resolve fine surface textures. For the tactile perception task of dexterous grasping, an NI-skin (48 × 48 elements; 69 units $cm^{-2}$) was integrated onto a commercial robotic hand (Inspire-Robots RH56DFX-R) to capture tactile information during object grasping. For tactile perception on large surfaces, a large-area NI-skin (96 × 96 elements; 30 units $cm^{-2}$) was attached to the outer surface of a humanoid robot (Unitree Robotics G1) to sense interactions with different contacts and touch gestures. For the fingertip fine-perception and nociception tasks, an NI-skin (24 × 24 elements; 156 units $cm^{-2}$) was mounted on the soft finger of a robotic gripper[50,51], which was installed on a commercial robotic arm (Universal Robots UR10e). Tactile signals were transmitted via a data bus using the USB or Transmission Control Protocol to a laptop computer serving as the central processor, which in turn controlled the robotic arm and gripper motion through Transmission Control Protocol/Internet Protocol communication. The interactive user software for the host computer was developed in Python 3.9 with the PyQt5 (including QtSerialPort), PyQtGraph, OpenCV and Pyserial frameworks. It supported real-time data transmission, waveform visualization, graphical rendering, tactile map visualization and data storage in HDF5 format (h5py).

### Construction of dataset

Grasping dataset: this dataset recorded tactile pressure data from the robotic hand's NI-skin during the grasping of ten object categories, with each object grasped four times. Data were collected at 100 Hz, with a grasp duration of 3–5 s, covering tactile responses from ten daily items (Fig. 4d). Touch dataset: this dataset captured dynamic pressure data from ten different touch gestures applied to the large-area NI-skin on the robotic body. Each gesture lasted 3–10 s and was performed by four subjects (age range: 21–28 years, 1 female and 3 male individuals), with data being recorded at 100 Hz (Fig. 4f). Pain-perception dataset: this dataset recorded tactile responses from the NI-skins on robotic fingers under five different sharpness levels (controlled by a power-law index), defined as varying pain intensity (0–100%; Supplementary Fig. 35). Data were collected at 100 Hz under a constant force of ~500 mN, with each pain level being recorded for 3–5 s. All datasets were stored in HDF5 format using h5py. All participants provided informed consent before participation in the study. All human–robot tactile interaction experiments were approved by the Ethics Committee of Southern University of Science and Technology (no. 2022DZX248).

### Training and optimization of the model

A 3D-DWT compression algorithm was used to reduce tactile data volume before network reasoning. The 3D tactile frames are decomposed into multilayer frequency-domain coefficients via wavelet transformation (bior1.3, two-level), separating primary features from fine details. The coefficients are then quantized by a quantization factor $Q$, converting them into integers to enhance sparsity through controllable precision loss. Finally, zlib entropy coding is applied to further minimize data volume (Supplementary Fig. 24). Decompression follows the reverse process of entropy decoding, dequantization and wavelet reconstruction.

For the grasping and touch classification tasks, a lightweight neural network 3D-CNN with a temporal attention head was developed (detailed architecture in Supplementary Fig. 26). The network was trained on the reconstructed tactile frames with cross-entropy loss using the Adam optimizer (learning rate $5 \times 10^{-4}$ for grasping, $1 \times 10^{-3}$ for touch, weight decay 0.01 and batch size 20). For pain-intensity regression, a generalized linear model with the binomial family and logit link was fitted on the three handcrafted features. All models were implemented in PyTorch (Python 3.9) on four NVIDIA T4 Tensor Core graphics processing units.

### Reporting summary

Further information on research design is available in the Nature Portfolio Reporting Summary linked to this article.

## Data availability

All data required to support the conclusions of this study are provided within the Article and its Supplementary Information. In addition, the

tactile interaction datasets generated in this study are publicly available via GitHub at https://github.com/jioy/NI-skins.git (ref. 52).

## Code availability

The codes that support the sensor data visualization, model training and interactive control are available via GitHub at https://github.com/jioy/NI-skins.git (ref. 52).

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## Author contributions

C.F.G., Z. Li, T.Wang, C.S. and B.D conceived the idea. B.D., J.S., Z. Li, Z. Liao and C.F.G. designed and fabricated the sensor array. Z. Li, W.S., B.L. and T.Wang designed the sensor readout architecture and the algorithm. Z. Li, B.D., H.H., Z.Liao, T. Wu, X.H., L.B. and H.K. designed the robotic system and conducted demonstrations. Z. Li, B.D. and J.S. wrote the paper. C.F.G., Z. Li, B.D., J.S., T.Wang and C.S. reviewed and rewrote the paper. C.F.G., T. Wang, C.S. and H.K. supervised the study.

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## Competing interests

The authors declare no competing interests.

## Additional information

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**Supplementary information** The online version contains supplementary material available at https://doi.org/10.1038/s42256-026-01325-6.

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**Extended Data Table 1 | Comparison of the NI-skin and representative sensory systems in terms of sensor density, total number of elements, readout architecture, readout rate, and sensor deployment**

| | Sensing mechanism | Sensor density (units $cm^{-2}$) | Total number of elements | Readout architecture | Readout rate (Hz) | Sensory modality | Deployment |
|---|---|---|---|---|---|---|---|
| Human skin[5–8] | Mechanoreceptors | ~ 240 | ~ 200,000 | Biological neural network | ~250 | Multimodal (Pressure, temperature, pain, friction ...) | Whole-body coverage |
| Kim, *Sci. Robot.*, (2022)[9] | Resistive sensors | 132 | 529 | Spike encoding | ~200 | Pressure | Robotic fingertip skin |
| Lee, *Sci. Robot.*, (2019)[27] | Resistive sensors | ~100 | 240 | Spike encoding | 1,000 | Pressure, temperature | Robotic fingertip skin |
| Liu, *Sci. Rob.* (2022)[48] | Synaptic transistors | ~ 50 | 168 | Spike encoding | ~5 | Pressure, pain | Robotic hand skin |
| Li, *Natl. Sci. Rev.*, (2026)[12] | Iontronic sensors | ~3 | 843 | Frequency encoding | 93 | Pressure, temperature | Robotic Arm coverage |
| Long, *Sci. Adv.*, (2023)[34] | Resistive sensors | 1 | 25 | Frequency encoding | 64 | Pressure | Robotic hand skin |
| Oh, *Sci. Adv.*, (2020)[16] | Thin-film transistors | 1,000 | 256 | Multiplexing array | 100 | Pressure | Robotic fingertip skin |
| He, *Adv. Mater.*, (2026)[13] | Piezoelectric and triboelectric sensors | 226 | ~200 | Multiplexing array | 20 | Pressure, contact height | Robotic fingertip skin |
| Zhao, *ACS Nano*, (2022)[26] | Thin-film transistors | ~123 | 4,096 | Multiplexing array | 50 | Pressure | Robotic hand skin |
| Sundaram, *Nature*, (2019)[21] | Resistive sensors | ~10 | 548 | Multiplexing array | 7.3 | Pressure | Wearable tactile glove |
| Shi, *Sci. Adv.*, (2023)[22] | Iontronic sensors | 8 | 784 | Multiplexing array | ~40 | Pressure | Robotic hand skin |
| Luo, *Nat. Electron.*, (2021)[23] | Resistive sensors | ~4 | >2,500 | Multiplexing array | 14 | Pressure | Large-area robotic coverage |
| Li, *Sci. Rob.*, (2020)[10] | Quadruple tactile sensors | ~1 | 10 | Wheatstone bridges | - | Pressure, temperature, thermal conductivity | Robotic hand skin |
| This work | Iontronic sensors | 1,110 | > 10,000 | Neural-hierarchical processing | 100 | Pressure, pain | Large-area robotic coverage |

The NI-skin combines high element density (1,110 units $cm^{-2}$), large number of sensors (>10,000 sensing units), and neural-hierarchical processing, enabling high-spatiotemporal-resolution sensing and large-area coverage of robots.

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| ☐ | ☒ A statement on whether measurements were taken from distinct samples or whether the same sample was measured repeatedly |
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| ☒ | ☐ A description of all covariates tested |
| ☒ | ☐ A description of any assumptions or corrections, such as tests of normality and adjustment for multiple comparisons |
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| ☒ | ☐ For Bayesian analysis, information on the choice of priors and Markov chain Monte Carlo settings |
| ☒ | ☐ For hierarchical and complex designs, identification of the appropriate level for tests and full reporting of outcomes |
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*Our web collection on statistics for biologists contains articles on many of the points above.*

## Software and code

Policy information about availability of computer code

| | |
|---|---|
| Data collection | Custom host software for real-time tactile data acquisition, visualization and storage was developed in Python 3.9 using PyQt5 (including QtSerialPort), PyQtGraph, OpenCV, pyserial and h5py. Data were stored in HDF5 format. The STM32F103V8T6 encoding controller firmware was developed using STM32CubeMX v6.14.0, STM32CubeIDE v1.18.0 and Keil MDK-ARM v5.39. FPGA-based parallel ADC acquisition and 512-point FFT demodulation were implemented on an Altera Cyclone IV E EP4CE10E22C8N FPGA, with the FPGA logic designed and compiled using Quartus II v13.0. |
| Data analysis | Data plotting and conventional data analysis were performed using Origin 2018 (OriginLab Corporation). Tactile-data processing and model development were performed in Python 3.9 using PyTorch. Frequency-encoded signals were decoded using a 512-point FFT. Spatiotemporal compression was implemented using a two-level 3D-DWT with the bior1.3 wavelet, coefficient quantization and zlib entropy coding. Grasped-object and touch-gesture recognition was performed using a lightweight 3D-CNN with a temporal attention head, trained using cross-entropy loss and the Adam optimizer. Pain-intensity prediction was performed using a generalized linear model with a binomial family and logit link. Model training was conducted on four NVIDIA T4 Tensor Core GPUs. |

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All manuscripts must include a data availability statement. This statement should provide the following information, where applicable:

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All data required to support the conclusions of this study are provided within the main text and Supplementary Information. Additionally, the tactile interaction datasets generated in this study are publicly available via GitHub at https://github.com/jioy/NI-skins.git

## Research involving human participants, their data, or biological material

Policy information about studies with human participants or human data. See also policy information about sex, gender (identity/presentation), and sexual orientation and race, ethnicity and racism.

| | |
|---|---|
| Reporting on sex and gender | Sex and gender were not variables of interest in this study and were not collected, because the human participants were involved only in providing predefined tactile interactions for evaluating the performance of the robotic tactile sensing system. No sex- or gender-dependent outcomes were investigated. |
| Reporting on race, ethnicity, or other socially relevant groupings | Race, ethnicity and other socially relevant groupings were not collected, as these characteristics were not relevant to the engineering objectives or outcome measures of this study. |
| Population characteristics | Four adult volunteers (age range: 21–28 years; 1 female and 3 males) were recruited by convenience sampling from the university community to perform predefined touch gestures. This recruitment approach may introduce self-selection bias; however, the participants were included only for evaluating the tactile sensing system rather than for population-level inference. Therefore, such bias is unlikely to affect the main conclusions of this study. |
| Recruitment | Participants were recruited on a voluntary basis from Southern University of Science and Technology. Participation was voluntary, and informed consent was obtained from all participants prior to data collection. |
| Ethics oversight | All human–robot tactile interaction experiments were approved by the Ethics Committee of Southern University of Science and Technology (No. 2022DZX248). Informed consent was obtained from all participants prior to participation in the study. |

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# Life sciences study design

All studies must disclose on these points even when the disclosure is negative.

| | |
|---|---|
| Sample size | Sample sizes for each dataset are as follows: Grasping Dataset, 10 object categories with 4 grasps per object (N=40); Touch Dataset, 10 gestures performed by 4 subjects (N=40); Pain-perception Dataset, tactile responses recorded under 5 sharpness levels (N=5 per level). |
| Data exclusions | No data were excluded from the analyses. |
| Replication | Each experimental condition was repeated as described above to ensure reproducibility: grasps and gestures were performed multiple times, and tactile responses were recorded for all trials. |
| Randomization | The order of dynamic touch gestures may be randomized to prevent sequence effects. Additionally, each participant repeats tasks multiple times to enhance data reliability and improve the model's accuracy. These steps help maintain the integrity of the study while allowing for meaningful analysis of human-robot interaction. |
| Blinding | Blinding was not applied in the study. |

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| ☒ | ☐ Palaeontology and archaeology |
| ☒ | ☐ Animals and other organisms |
| ☒ | ☐ Clinical data |
| ☒ | ☐ Dual use research of concern |
| ☒ | ☐ Plants |

## Methods

| n/a | Involved in the study |
|---|---|
| ☒ | ☐ ChIP-seq |
| ☒ | ☐ Flow cytometry |
| ☒ | ☐ MRI-based neuroimaging |

# Plants

Seed stocks

The study does not involve novel plant genotypes, seed stocks, authentication of plant materials, or any experiments related to plants. Its focus is exclusively on human-robot interaction, particularly the collection and analysis of tactile signals and the development of robotic perception and manipulation systems.

Novel plant genotypes

The study does not involve novel plant genotypes, seed stocks, authentication of plant materials, or any experiments related to plants.

Authentication

The study does not involve novel plant genotypes, seed stocks, authentication of plant materials, or any experiments related to plants.